A Statistical Analysis of Wildland and Prescribed Fires in the Changing California Climate

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Introduction

With climate change-driven large, destructive wildfires becoming more frequent, especially in California, using data to identify effective or ineffective prevention strategies, such as prescribed burns, can inform future prevention strategies. There exists a large body of research demonstrating the effectiveness of prescribed burns for mitigating the effects of large wildfires; however, much of this research was performed in the previous century and did for climate change's impact on prescribed burn efficacy. Additionally, the data needed for such an analysis is fragmented and complex, requiring the synthesis of wildfire, prescribed fire, and climate data from various sources. Statistical data science methods provide tools for making sense of the immense heterogenous data relevant to analyzing the relationship between prescribed fires, wildfires, and the climate. This research explores the following questions:

1. Are prescribed fires effective at reducing wildfire size?
2. Are prescribed burns effective at reducing the size of wildfires while controlling for climate-relevant variables?
3. How do these relationships change at different spatial-temporal scales?

Methods and Materials

In this work, we present an initial statistical analysis on a new dataset created from fusing prescribed fire and wildland fire records from The California Department of Forestry and Fire Protection (CalFire) [1, 2] with climate data from the National Oceanic and Atmospheric Administration (NOAA) [3]. A suite of statistical methods, including descriptive, inferential, and predictive methods are applied to the dataset to analyze if prescribed burns are effective in mitigating large wildfires in recent history while accounting for climate change. Descriptive methods are used to summarize the data trends, while inferential methods identify significant trends. Inferential methods include non-parametric tests, including the Mann-Whitney U test for differences in acres burned between predictor levels, and parametric tests, such as the χ^2 test for independence of region on number of fires. A time series analysis with cross correlation is used to identify which variables are related to wildfire acres burned at different time lag values. Additionally, a regression analysis is performed to identify significant predictors of wildfire acres burned, as well as to estimate expected acres burned when climate variables, such as average temperature and precipitation, change in accordance with climate models.

Results

Inferential tests find that wildfires with prescribed fires performed within 50 km and 12 months are smaller on average than wildfires with no prescribed burn treatment. When grouped across the entire state with a monthly time scale, results indicate prescribed burns do not impact the number of acres burned due to wildfire. However, when considering smaller regions, the impact of prescribed burns is measurable. At a state-wide level, the Palmer Drought Severity Index (PDSI), average temperature, maximum temperature, and precipitation are related to wildland fire acres burned, with increased

temperature, increased drought, and lower precipitation positively correlated with wildfire acres burned. When including climate variables and prescribed burn variables in a regression model, both variable types are significant predictors of wildfire size. Overall climate variables have a stronger impact on wildfire size, though prescribed burns are associated with slightly smaller wildfires.

Discussion and Conclusion

Prescribed burns are indeed associated with reduced wildfire sizes when considered alone and in conjunction with climate variables. Although, the impact of prescribed burns on wildfire size is low, especially when considering the impact of climate variables such as drought and precipitation. With climate change resulting in more extreme temperature maximums and droughts, prescribed burns may not be sufficient in mitigating large fires. Future work will investigate using integrated spatial-temporal models, as well as investigate more rigorous methods of casual inference for examining the impact of prescribed burns. Research into additional wildfire prevention methods is critical given the limited impact of prescribed burns.

References

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